



KIET
GROUP OF INSTITUTIONS
Connecting Life with Learning



A
Project Synopsis
on
iRetina: Automated Diabetic Retinopathy Detection Using
Deep Learning
submitted as partial fulfillment for the award of
BACHELOR OF TECHNOLOGY
DEGREE

SESSION 2025-26

in
Computer Science & Engineering

By

Ankit Kumar (Roll No.-2300290100052) (Team Leader)

Aditya Sharma (Roll No.-2300290100023)

Prakhar Verma (Roll No.- 2300290100177)

Paras Lahauriya (Roll No.- 2300290100169)

Under the supervision of

Dr. Himanshi Chaudhary

KIET Group of Institutions, Ghaziabad

Affiliated to

Dr. A.P.J. Abdul Kalam Technical University, Lucknow
(Formerly UPTU)

Academic Year 2025-26

Abstract

Diabetic Retinopathy (DR) is a common complication of diabetes mellitus, which causes lesions on the retina that effect vision. If it is not detected early, it can lead to blindness. Unfortunately, DR is not a reversible process, and treatment only sustains vision. DR early detection and treatment can significantly reduce the risk of vision loss. The manual diagnosis process of DR retina fundus images by ophthalmologists is time-, effort-, and costconsuming and prone to misdiagnosis unlike computer-aided diagnosis systems. Recently, deep learning has become one of the most common techniques that has achieved better performance in many areas, especially in medical image analysis and classification. Convolutional neural networks are more widely used as a deep learning method in medical image analysis and they are highly effective. For this article, the recent state-of-the-art methods of DR color fundus images detection and classification using deep learning techniques have been reviewed and analyzed. Furthermore, the DR available datasets for the color fundus retina have been reviewed. Difference challenging issues that require more investigation are also discussed.

Introduction

In the healthcare field, the treatment of diseases is more effective when detected at an early stage. Diabetes is a disease that increases the amount of glucose in the blood caused by a lack of insulin. It affects 425 million adults worldwide . Diabetes affects the retina, heart, nerves, and kidneys. Diabetic Retinopathy (DR) is a complication of diabetes that causes the blood vessels of the retina to swell and to leak fluids and blood. DR can lead to a loss of vision if it is in an advanced stage. Worldwide, DR causes 2.6% of blindness. The possibility of DR presence increases for diabetes patients who suffer from the disease for a long period. Retina regular screening is essential for diabetes patients to diagnose and to treat DR at an early stage to avoid the risk of blindness. DR is detected by the appearance of different types of lesions on a retina image. These lesions are microaneurysms (MA), haemorrhages (HM), soft and hard exudates .

- Microaneurysms (MA) is the earliest sign of DR that appears as small red round dots on the retina due to the weakness of the vessel's walls. The size is less than 125 μm and there are sharp margins. Michael et al. classified MA into six types. The types of MA were seen with AOSLO reflectance and conventional fluorescein imaging.
- Haemorrhages (HM) appear as larger spots on the retina, where its size is greater than 125 μm with an irregular margin. There are two types of HM, which are flame (superficial HM) and blot (deeper HM).
- Hard exudates appear as bright-yellow spots on the retina caused by leakage of plasma. They have sharp margins and can be found in the retina's outer layers.
- Soft exudates (also called cotton wool) appear as white spots on the retina caused by the swelling of the nerve fiber. The shape is oval or round. Red lesions are MA and

HM, while bright lesions are soft and hard exudates (EX). There are five stages of DR depending on the presence of these lesions, namely, no DR, mild DR, moderate DR, severe DR and proliferative DR.

The automated methods for DR detection are cost and time saving and are more efficient than a manual diagnosis. A manual diagnosis is prone to misdiagnosis and requires more effort than automatic methods. This paper reviews the recent DR automated methods that use deep learning to detect and to classify DR.

- Motivation - The rapid increase in diabetic patients worldwide has made diabetic retinopathy a major public-health concern. Rural and semi-urban areas face a shortage of trained ophthalmologists, making timely diagnosis difficult. Manual screening is also slow and prone to fatigue-induced inconsistencies. With the growing shift toward telemedicine and automated hospital workflows, there is a strong need for AI-driven tools that can support large-scale screening. Affordable deep-learning solutions have the potential to significantly reduce preventable blindness by enabling early and accurate detection.
- Problem Gaps Identified - Current manual diagnosis methods cannot scale to meet the growing screening demands. Available retinal datasets often suffer from noise, uneven illumination, and inconsistent image quality, reducing model reliability. Many existing AI models classify only normal vs. abnormal cases and fail to detect all five stages of diabetic retinopathy. Additionally, very few real-time systems are optimized for deployment in low-resource environments, limiting their practical usability in rural healthcare centers.
- Proposed Contribution - The iRetina system aims to deliver an end-to-end automated pipeline for diabetic retinopathy detection using deep learning. The workflow integrates advanced preprocessing techniques such as cropping, normalization, CLAHE, and denoising to enhance image quality. Convolutional Neural Networks—such as VGG16, ResNet50, and Inception-V3—will be trained for five-stage DR classification using transfer learning and extensive data augmentation. The system also includes a lightweight web-based interface for easy access and adds explainability through heatmap-based lesion localization using Grad-CAM.
- Technology Stack - The project leverages TensorFlow/Keras or PyTorch for deep learning, using architectures like ResNet50, Inception-V3, and DenseNet121. Image preprocessing is carried out using OpenCV and NumPy. Model training utilizes public datasets such as Kaggle EyePACS, DIARETDB1, and Messidor. A Flask or FastAPI backend powers the deployed prototype, integrated with the iRetina web interface, while training is performed on GPU-enabled hardware for efficient computation.

Literature Survey

The study by Wejdan L. Alyoubi et al. (2020) provides a comprehensive review of state-of-the-art deep learning techniques for diabetic retinopathy detection. DR lesions such as microaneurysms, hemorrhages, and exudates form the core diagnostic features, and the authors highlight the importance of CNN architectures in extracting these patterns. Traditional machine learning methods require manual feature extraction, whereas deep learning automatically learns discriminative features directly from fundus images.

The authors outline several publicly available datasets—Kaggle EyePACS, DIARETDB1, DRIVE, HRF, IO-DRI, Messidor, and IDRiD—each with varying resolutions and annotation depth. They emphasize that dataset quality is essential because CNNs require large, diverse, and balanced datasets to outperform human graders. Many studies preprocess images using normalization, resizing, contrast enhancement, and denoising to maintain consistency.

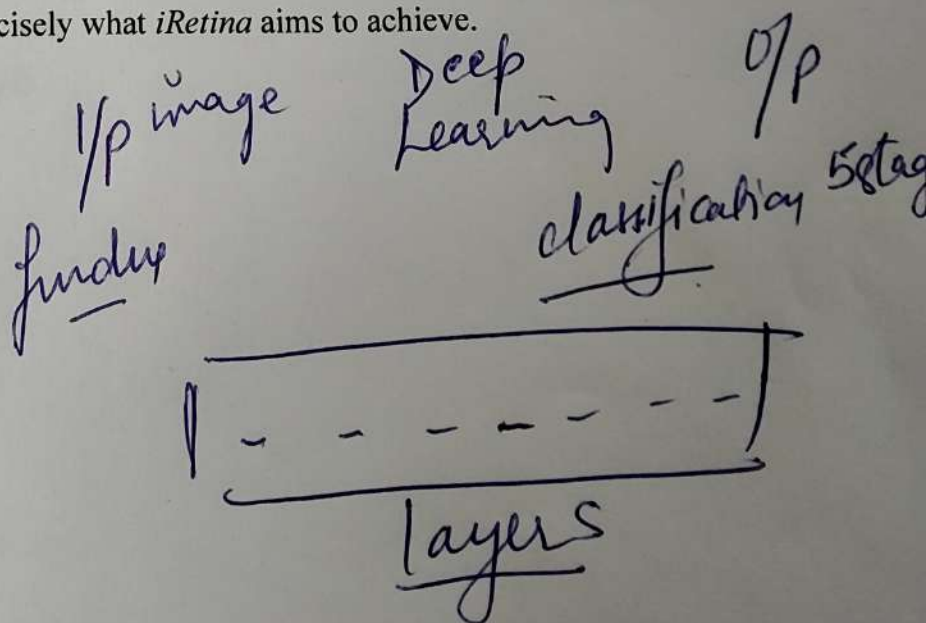
Multiple CNN-based models such as AlexNet, VGG16, Inception-V3, and ResNet have been used in DR classification tasks. Transfer learning is particularly effective, enabling faster training and improved accuracy even with limited data. Some studies focus on binary classification (DR vs Non-DR), while others target multi-class classification for 5 DR stages. Several works integrate lesion detection (microaneurysms, hard exudates) using patch-based CNNs, improving interpretability. However, most systems either detect lesions or classify stages, but not both.

Recent advancements include fully convolutional networks for vessel segmentation and hybrid CNN-Random Forest models for lesion identification. Despite high accuracy results (often >90%), challenges remain in poor-quality image handling, incorrect labeling, and limited generalizability across datasets. The literature suggests strong potential for AI-enabled DR screening, but highlights the need for robust, end-to-end systems that integrate lesion detection and stage classification—precisely what *iRetina* aims to achieve.

Methodology/ Planning of work

Step 1: Dataset Collection

- Kaggle EyePACS dataset
- DIARETDB1
- Messidor dataset



Step 2: Image Preprocessing

- Resizing to fixed dimensions
- Cropping the circular fundus region
- Green channel extraction (for better contrast)
- CLAHE contrast enhancement
- Noise removal (Gaussian/Median filtering)
- Data augmentation (rotation, flipping, brightness shift)

Step 3: Model Development

- Selection of pretrained CNN (Inception-V3 / ResNet50 / DenseNet)
- Fine-tuning last layers
- Training for 5-class classification (0–4 DR stage)

Step 4: Evaluation

- Accuracy, Precision, Recall, F1 Score, AUC metrics
- Confusion matrix
- Grad-CAM visualization for explainability

Step 5: Deployment (Prototype)

- Flask/FastAPI backend
- Integration with iRetina front-end web interface
- Upload → Predict → Display stage & lesion heatmap

Facilities required for proposed work

Software Requirements

- Python 3.9+
- TensorFlow / Keras / PyTorch
- OpenCV
- NumPy, Pandas, Matplotlib
- Flask/FastAPI for deployment

- GPU-enabled Jupyter Notebook/Google Colab

Hardware Requirements

- GPU-based training machine (NVIDIA RTX 3060 or above recommended)
- 8–16 GB RAM
- High-resolution display for fundus image visualization

References

[1] W. L. Alyoubi, W. M. Shalash, and M. F. Abulkhair, "Diabetic retinopathy detection through deep learning techniques: A review," *Informatics in Medicine Unlocked*, vol. 20, 2020.

Diabetic retinopathy detection ...

[2] Krizhevsky, A., Sutskever, I., & Hinton, G. E. "ImageNet classification with deep convolutional neural networks," *Communications of the ACM*, 2017.

[3] Szegedy, C., et al. "Rethinking the inception architecture for computer vision," *CVPR*, 2016.

[4] He, K., et al. "Deep residual learning for image recognition," *CVPR*, 2016.

[5] Kaggle, "Diabetic Retinopathy Detection Dataset," 2015.